Fraction of Vacant Lattice Points in FCC Palladium

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12. The density of a sample of FCC palladium is $11.98\ \mathrm{g/cm^3}$***, and its lattice parameter is*** 3.8902 Å. ***Calculate: a)*** *The fraction of the lattice points that contain vacancies. (Solution:* 0.00204*)* ***b)*** *The total number of vacancies in a cubic centimeter of* Pd. *(Solution:* $1.39 \times 10^{20}\ \mathrm{vacancies/cm^3}$*)* ***Data:*** $M_{\mathrm{Pd}} = 106.4\ \mathrm{g/mol}$*;* $N_A = 6.022 \times 10^{23}\ \mathrm{atoms/mol}$.

Solution: 0.00204

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Material: Metallic palladium (Pd).
- Crystal structure: Face-Centred Cubic (FCC) $\implies n_{\mathrm{sites}} = 4$ lattice sites/cell.
- Measured lattice parameter: $a = 3.8902\ \text{Å} = 3.8902 \times 10^{-8}\ \mathrm{cm}$.
- Measured real density: $\rho_{\mathrm{real}} = 11.98\ \mathrm{g/cm^3}$.
- Molar atomic mass: $M_{\mathrm{Pd}} = 106.4\ \mathrm{g/mol}$.
- Avogadro number: $N_A = 6.022 \times 10^{23}\ \mathrm{atoms/mol}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Ideal Theoretical Density of the Perfect Crystal (ρ_{ideal}) [Session 4 Slides 7, 16]:**

If all lattice points were occupied by palladium atoms with no vacancies at all:

$$\rho_{\mathrm{ideal}} = \frac{n \cdot M_{\mathrm{Pd}}}{a^3 \cdot N_A} = \frac{4 \cdot M_{\mathrm{Pd}}}{a^3 \cdot N_A}$$

2. **Fraction of Vacant Lattice Points (f_v):**

The existence of vacancies reduces the mass contained in the cell without appreciably altering its lattice parameter a. The vacancy fraction is the relative density deficit:

$$f_v = \frac{\rho_{\text{ideal}} - \rho_{\text{real}}}{\rho_{\text{ideal}}} = 1 - \frac{\rho_{\text{real}}}{\rho_{\text{ideal}}}$$

3. **Total Number of Vacancies per Cubic Centimetre (n_v):**

The total number density of lattice points per cubic centimetre is:

$$N = \frac{n_{\text{sites}}}{V_C} = \frac{4}{a^3}$$

The absolute number of vacancies per cubic centimetre is:

$$n_v = f_v \cdot N$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Ideal Theoretical Density (ρ_{ideal}):

- Unit cell volume:

$$V_C = a^3 = (3.8902 \times 10^{-8}\ \text{cm})^3 = 5.88729 \times 10^{-23}\ \text{cm}^3$$

- Theoretical mass of the perfect unit cell:

$$m_{\text{cell}} = \frac{4 \times 106.4\ \text{g/mol}}{6.022 \times 10^{23}\ \text{mol}^{-1}} = \frac{425.6}{6.022 \times 10^{23}} = 7.06742 \times 10^{-22}\ \text{g}$$

- Theoretical density:

$$\rho_{\text{ideal}} = \frac{7.06742 \times 10^{-22}\ \text{g}}{5.88729 \times 10^{-23}\ \text{cm}^3} = \mathbf{12.0045}\ \text{g/cm}^\mathbf{3}$$

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2. Part a: Fraction of Vacant Lattice Points (f_v):

$$f_v = \frac{\rho_{\text{ideal}} - \rho_{\text{real}}}{\rho_{\text{ideal}}} = \frac{12.0045 - 11.9800}{12.0045} = \frac{0.0245}{12.0045}$$

$$f_v = \mathbf{0.00204} \quad (\approx 0.204\%)$$

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3. Part b: Number of Vacancies per Cubic Centimetre (n_v):

- Total concentration of lattice points (N):

$$N = \frac{4}{a^3} = \frac{4}{5.88729 \times 10^{-23}\ \text{cm}^3} = 6.7943 \times 10^{22}\ \text{sites/cm}^3$$

- Volumetric vacancy concentration:

$$n_v = f_v \cdot N = (0.002043) \times (6.7943 \times 10^{22}\ \text{cm}^{-3}) = \mathbf{1.388 \times 10^{20}}\ \text{vacancies/cm}^3 \approx \mathbf{1.39 \times 10^{20}}\ \text{vac/cm}^3$$

The unrounded fraction $f_v = 0.0020432$ gives $n_v = 1.388 \times 10^{20}\ \text{cm}^{-3}$; both results agree with the official 0.00204 and $1.39 \times 10^{20}\ \text{cm}^{-3}$.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Pycnometric and Dilatometric Detection:** This problem reflects the classic Simmons and Balluffi method: by comparing the lattice-parameter variation obtained by X-ray diffraction ($\Delta a/a$) with the macroscopic dimensional variation from dilatometry ($\Delta L/L$), or by contrasting the hydrostatic density with the X-ray theoretical one, the absolute concentration of thermal vacancies in noble metals such as palladium, platinum and gold is measured non-destructively.

Study notes written from the official course material (Problems T2_CrystStruct.pdf; Problems T2 defects.pdf).
Online version: subjects/aerospace-materials-1/problemas/topic-2-problems.html